

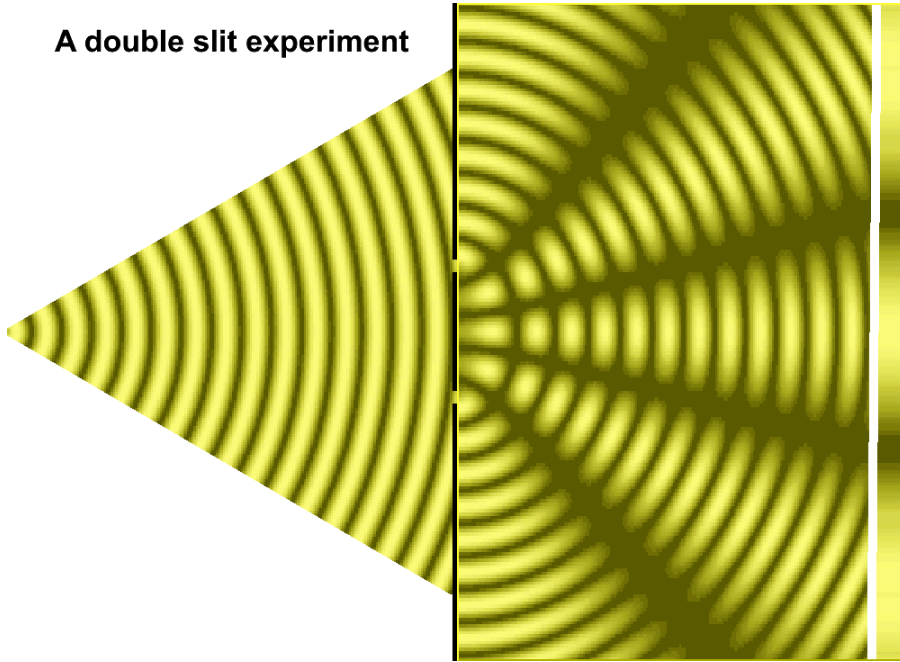
23 popquiz resolution interference, S and P

 This is a preview of the draft version of the quiz

Started: Mar 23 at 6:55pm

Quiz Instructions

A double slit experiment



Question 1 17 pts

https://phet.colorado.edu/sims/html/wave-interference/latest/wave-interference_all.html 
(https://phet.colorado.edu/sims/html/wave-interference/latest/wave-interference_all.html)

Which wavelength (color) will diffract less for a given opening?

(that is which wavelength (color) spreads out the least after going through the small hole)



yellow



red



blue

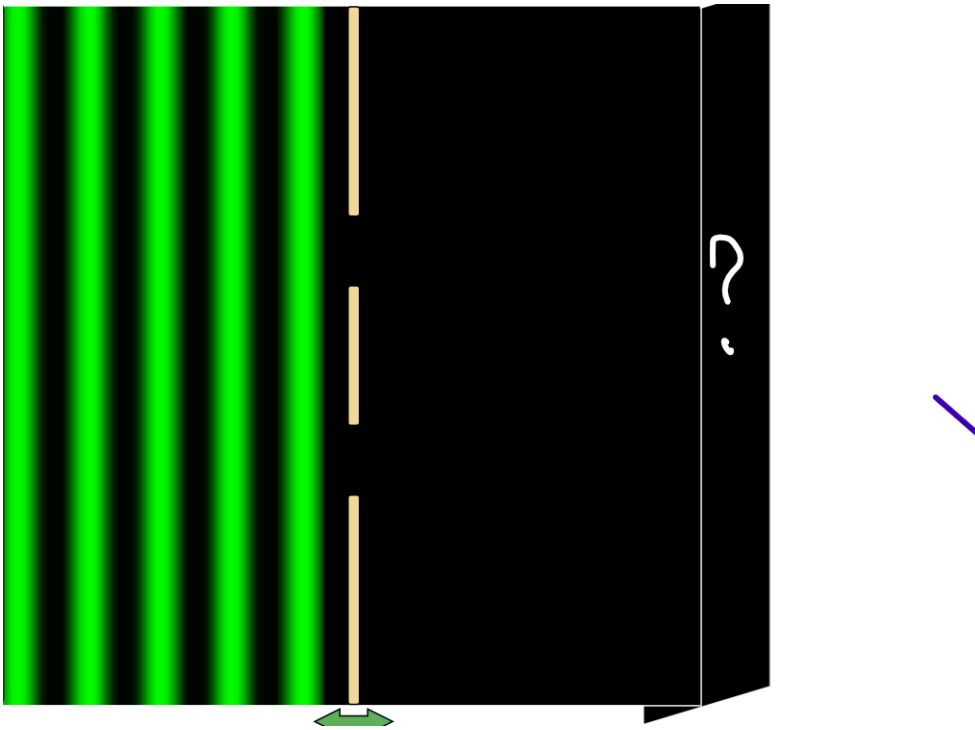


green



Question 2 17 pts

With the famous 2 slits experiment, what pattern is seen on a screen.



if you missed the class:

https://phet.colorado.edu/sims/html/wave-interference/latest/wave-interference_all.html

(https://phet.colorado.edu/sims/html/wave-interference/latest/wave-interference_all.html)

select slits then select 2 slits and screen. Turn on the laser (you see plane waves)

- ☐ a serie of bright and dark bands
- ☐ only 2 bright bands corresponding to the 2 slots)
- ☐ a large bright spot
- ☐ The screen is all dark



Question 3 17 pts

What is the resolution of a telescope/microscope? hint;



- ☐ its ability to magnify an object.
- ☐ its ability to distinguish two adjacent objects close together in the sky
- ☐ its ability to focus more than just visible light for imaging
- ☐ its ability to make distant objects appear much closer to us



Question 4 17 pts

a IR is radiating in 3D. Its power is 40W. What is the amount of energy per second going through a unit area at a distance $d = 50\text{cm}$ from the source ? (find S in W/m^2)

hint: convert cm to m first. slide 26 gives you the equation (bold)

- ☐ about $13\text{W}/\text{m}^2$
- ☐ anout $40\text{W} / \text{m}^2$
- ☐ about $126\text{W}/\text{m}^2$
- ☐ about $8\text{W}/\text{m}^2$



Question 5 17 pts

A EM wave has a wavelength of $5 \cdot 10^{-8} \text{ m}$. First convert to nm (nanometer) then use my slide 15

Whats this EM wave?

$1 \text{ nm} = 10^{-9} \text{ m}$

☐

microwaves

☐

gamma rays

☐

UV

☐

IR



Question 6 17 pts

violet has a wavelength of about 400nm. frequency is about

☐about $7.5 \cdot 10^{14}$ Hz☐about $1.3 \cdot 10^{12}$ Hz☐about $1.3 \cdot 10^{-15}$ Hz☐about $1.3 \cdot 10^{-6}$ Hz

Not saved

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